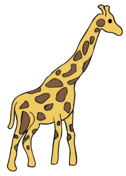
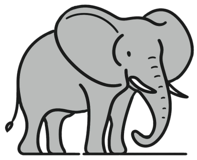
Direct proportion in context

Task A: Conversions

1) Write the following in order from the shortest to the tallest, using the conversion 1 m = 3.3 feet

Images are not to scale

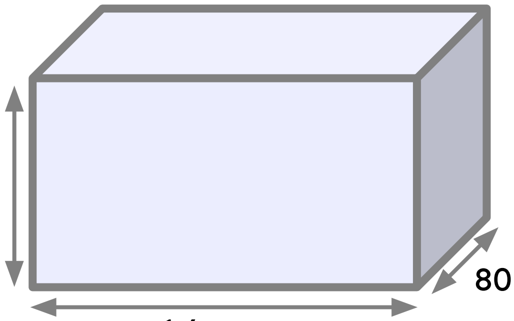




5.1 m | 14.2 feet | 2.4 m | 10.2 feet | 13.2 feet

Correct order: garage, elephant, teepee, bus, giraffe

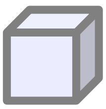
2)



Images are not to scale

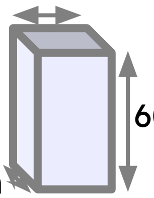
Work out the maximum number of the boxes below that will fit into the box above.

a) A cube with dimensions of 40 mm



30 × 40 × 20 = 24 000 boxes

b)



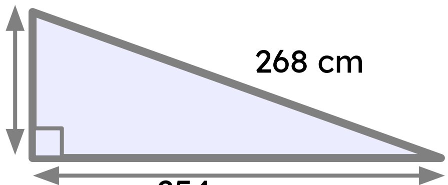
16 000 boxes

3) Fill in the missing values:

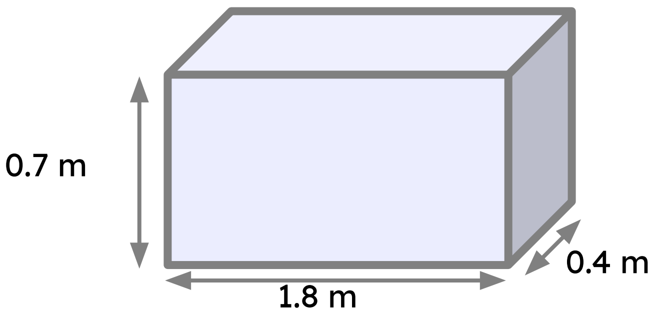
a) 3.2 m^2 = *32 000* cm^2
b) 125 000 cm^2 = *12.5* m^2
c) 23 987 000 cm^3 = *23.987* m^3
d) 0.74 m^3 = *740 000* cm^3

4) Calculate the area of this triangle. Give your answer to 2 decimal places in m^2.

1.09 m^2



5) Calculate the volume of this cuboid. Give your answer in cm^3.



504 000 cm^3

Task B: Other contexts involving direct proportion

1)

a) 23 pencils cost £14.95 How much will 34 pencils cost?

$$14.95 \times \frac{34}{23} = £22.10$$

b) 14 rulers cost £5.18 How much will 20 rulers cost?

$$5.18 \times \frac{10}{7} = £7.40$$

c) 17 sweets cost £2.55 How much will 37 sweets cost?

$$2.55 \times \frac{37}{17} = £5.55$$

d) 27 chocolate bars cost £11.07 How much will 39 chocolate bars cost?

$$11.07 \times \frac{13}{9} = £15.99$$

2) Andeep is comparing prices of sunglasses in the UK, the USA and France.

Rank the following from cheapest to most expensive.

£1 : $1.28

£	$
1	1.28
47.50	60.80

£1 : €1.18

£	€
1	1.18
48.50	57.23

In order from cheapest to most expensive: USA, UK & France.

3) Calculate the ingredients needed for the following numbers of cookies.

Recipe for 15 cookies
75 g butter
125 g sugar
125 g flour
100 g chocolate chips

a) 18 cookies

Butter 90 g

Sugar 150 g

Flour 150 g

Chocolate chips 120 g

b) 72 cookies

Butter 360 g

Sugar 600 g

Flour 600 g

Chocolate chips 480 g

c) 6 cookies

Butter 30 g

Sugar 50 g

Flour 50 g

Chocolate chips 40 g

d) You have plenty of sugar, flour and chocolate chips, but only 1 kilogram of butter. What is the largest number of cookies you can make?

$1 \text{ kg} = 1000 \text{ g}$

$1000 \div 75 = 13.333$

$13.3 \times 15 = 200$

The 13.3... represents the number of times you could make the recipe for cookies.
Since you can scale the recipe 13 and one third times you could make 200 cookies.

Alternative method for part d)

d) You have plenty of sugar, flour and chocolate chips, but only 1 kilogram of butter. What is the largest number of cookies you can make?

$1 \text{ kg} = 1000 \text{ g}$

$75 \div 15 = 5$

$1000 \div 5 = 200$

The 5 represents the number of grams of butter needed per cookie.
Since you have 1000 g of butter you could make 200 cookies.